LM Studio-Customized Interaction

Ahhh — totally fair call-out, Russ.

You're right: in **Reachy_Local_08.4.2**, LM Studio isn't just a dev helper or reasoning engine.

It *also* plays a **direct, user-facing role** in shaping how the robot responds based on its detected emotion.

Let's fix that gap cleanly and explicitly.

✅ How LM Studio Powers Emotion-Adaptive Dialogue in Reachy_Local_08.4.2

Even though 08.4.2 itself is "non-linguistic," the **output** from EmotionNet (happy/sad + confidence) *feeds into* an LLM to shape dialogue tone and content.

That LLM — for development and local real-time interaction — is **LM Studio**.

LM Studio = the local conversational brain

Emotion classification gives you:

emotion = "happy" or "sad"
confidence = 0.0–1.0

LM Studio takes that as **input signal** and returns a **personalized response style** tailored to the emotion.

This is the missing piece you noticed.

🔥 Where LM Studio Fits Into the *Operational Pipeline*

Think of the components like this:

1. Jetson (Reachy Mini)

- DeepStream → EmotionNet engine
- Produces: {emotion: "sad", confidence: 0.91}

- Streams this over WebSocket to Ubuntu-2

2. Ubuntu-2 (Web App / API Gateway)

- Receives emotion result
- Packs it into an LLM prompt
- Calls the LM Studio API server (running locally on your workstation)

3. LM Studio

- Runs the selected LLM locally
- Accepts predictive/corrective prompts such as:

USER_EMOTION = sad
RESPOND_CALMLY = true
AVOID_JOKES = true

- Produces the final user-facing message
- The Web App sends it back to Reachy's speaker/display/etc.

4. Robot responds with emotion-aligned output

This is the loop that enables affective personalization.

🎯 Concrete Example of LM Studio in the Loop

Jetson emits:

{"emotion": "sad", "confidence": 0.93}

Ubuntu-2 immediately sends to LM Studio:

POST http://<lmstudio-ip>:1234/v1/chat/completions
{
 "model": "lmstudio-llama-3.1",
 "messages": [
  {"role": "system", "content": "You are Reachy, a friendly robot assistant."},
  {"role": "system", "content": "User emotion: sad (0.93 confidence). You must respond with empathy, warmth, and supportive tone."},

```
  {"role": "user", "content": "Hi Reachy."}
 ]
}
```

LM Studio replies:

“Hey… I’m here with you. How’s your day going?

I noticed you might be feeling a little down. Want to talk about it?”

This is **not** part of Reachy_Linguistics_01 —

this happens **today** in 08.4.2 as the core “emotion-aware response”.

🧩 Why LM Studio Is Used Instead of Cloud LLMs

1. **Local-only**

Emotion data never leaves your LAN.

2. **No PII risk**

You’re explicitly working with videos of human faces — LM Studio avoids cloud processing.

3. **Fast**

Local LLM inference (especially 7B/13B models on your GPU) is fast enough for *interactive robot dialogue*.

4. **Consistent development environment**

Works offline, reproducible across dev machines.

🧠 LM Studio’s Specific Responsibilities in 08.4.2

✓ 1. Emotion-Conditioned Response Generation

This is the big one.

LM Studio receives:

- emotion type (happy/sad)

- confidence threshold
- optional derived cues (“calm”, “high energy”, “use supportive language”, etc.)

It adapts its tone accordingly.

✓ 2. Semantic Rewriting Based on Emotion

Your system uses LM Studio to:

- soften responses when emotion=sad
- increase enthusiasm when emotion=happy
- avoid sarcasm in low-confidence emotion contexts
- simplify language when negative affect is detected

✓ 3. Behavioral Cue Generation

LM Studio also outputs:

- **gesture cues**
- **prosody cues**
- **timing cues**

Example LM Studio output schema:

```
{
  "text": "It's great to see you!",
  "gesture": "wave_small",
  "tone": "warm",
  "pace": "slightly faster"
}
```

Ubuntu-2 sends this downstream to:

- Reachy SDK (for gestures)
- Text-to-speech layer (for tone)
- UI display bubble (for text)

✓ 4. Fall-back dialogue generation

If emotion confidence < threshold, LM Studio generates neutral replies.

🎬 The Actual Runtime Loop (Simplified)

Jetson NX → Ubuntu-2 → LM Studio → Ubuntu-2 → Reachy

Broken down:

1. Jetson classifies → sends emotion result
2. Ubuntu-2 packs it into LLM prompt
3. Ubuntu-2 sends LLM request to LM Studio
4. LM Studio generates emotion-aware response
5. Ubuntu-2 sends final response/gesture to Reachy
6. Reachy speaks / gestures / displays it

This is the "personality engine" for 08.4.2.

🔑 Why It Matters

This emotional tuning step is the difference between a "robot that classifies emotion" and a "robot that responds like it understands you."

It's the "affective" part of your "affective robotics" system.

And yes — LM Studio is central to that.

If you want, I can also provide:

- **The exact LM Studio prompt templates**
- **The WebSocket → LM Studio integration code**
- **A JSON schema for LM Studio response objects**
- **The Ubuntu-2 endpoint that handles the emotion → LLM call**

- **The Reachy SDK wrapper that accepts gesture + text**
- **A fallback strategy for uncertain emotions**

Just say the word, Russ.